Muhammad Arslan

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TECHNICAL SKILLS

- **Languages:** JavaScript, Java, C++, Python, .NET framework
- **Frontend:** React.js, HTML5, CSS3, Tailwind CSS, Flexbox/Grid
- **Backend & Cloud:** Node.js, Express.js, Java Spring Boot, Firebase
- **Databases:** MongoDB (NoSQL), SQL Server
- **Tools & Testing:** Git/GitHub, LoadRunner, JUnit, REST APIs
- **AI & Data:** Prompt Engineering, Data Mining

PROFESSIONAL EXPERIENCE

AI-Assisted Web and Agentic AI Intern July 2026 - Sep 2026

Remote Internship Arbisoft Ltd.

- Completed an eight-week AI-focused remote internship covering full-stack web development, REST APIs, authentication, testing and agentic AI.
- Applied AI-assisted coding, prompt engineering and code review; reviewed generated implementations and unit and integration tests with mentor feedback.
- Developed and demonstrated CampusGo through the self-directed project phase, from proposal and architecture to implementation, testing and technical documentation.

Front-End Developer July 2025 - Aug 2025

Internship Nexum

- Built 3+ React-based applications including a movie search platform and AI text summarizer using REST APIs.
- Implemented React Hooks, component-based architecture, and conditional rendering to improve UI responsiveness.
- Integrated external APIs and handled asynchronous operations with proper error handling.

PROJECTS

CampusGo / Pick-and-Drop System July 2026 - Sep 2026

React, TypeScript, Python, FastAPI, PostgreSQL/PostGIS, LangGraph, Ollama

- Built a full-stack student ride coordination platform with identity verification, ride requests and seat offers, route matching, trip groups and emergency escalation.
- Developed REST APIs using FastAPI, asynchronous SQLAlchemy and Pydantic in a layered architecture; implemented JWT authentication and Argon2id password hashing.
- Implemented geospatial queries with PostGIS and OSRM routing, OpenCV-based student verification, and deterministic eligibility, seat-capacity and trip-state validation.
- Integrated LangGraph multi-agent workflows with local Ollama LLMs for ride assistance, replacement-rider coordination and emergency strategy, with schema validation and tool-call guardrails.
- Added durable notifications, retry and deduplication logic, Docker Compose configuration and Alembic migrations; validated workflows with backend unit/integration tests and frontend Playwright tests.

Eventer MERN Stack

- Designed and developed a full-stack event management system enabling users to create, organize, and manage events with a scalable architecture supporting ~250–500 concurrent users.
- Implemented secure authentication and authorization using JSON Web Tokens (JWT), including role-based access control (RBAC) for admin and user-level permissions.
- Engineered efficient data workflows for event and user operations, facilitating seamless record creation, updates, and deletion while maintaining consistent real-time synchronization.
- Integrated MongoDB Atlas for cloud-based data storage, ensuring efficient querying and scalability for growing user activity.
- Developed notification features, including email and SMS integration, to send event reminders and enhance user engagement.

Book Linker ReactJS, JAVA Spring Boot, Firebase

- Architected a real-time book exchange platform using Spring Boot and Firebase, enabling seamless peer-to-peer transactions and donation workflows.
- Engineered a secure role-based access control (RBAC) system to manage user permissions, ensuring data integrity for book lending and exchange requests.
- Optimized frontend performance by implementing a component-driven architecture in React, which streamlined the UI development process and improved responsiveness.

SMS .NET Framework, C++

- Developed a desktop-based student management system to efficiently manage student records, including attendance, academic performance, and personal information.
- Implemented role-based functionality for Admin, Teacher, and Student, ensuring controlled access to system features and data.
- Designed modules enabling teachers to record attendance and assign marks, with real-time updates to student academic profiles.
- Enabled centralized data management for student and teacher records through optimized CRUD operations, ensuring data consistency and streamlined administrative workflows.

Fast Electrics .NET Framework, C++

- Desktop application designed to help manage electricity consumption
- Generate an optimized schedule derived from the device's daily, weekly, and monthly usage frequency.

CERTIFICATIONS AND PUBLICATIONS

- CSS - The Complete Guide 2022 (incl. Flexbox, Grid & Sass)
- The Beginner's Guide to React
- The Ultimate React Course

AWARDS & ACHIEVEMENTS

Web Hackathon University of Central Punjab (UCP)

3rd Position

- Competed against teams from various universities to develop a functional web solution within a limited timeframe.
- Awarded for excellence in UI/UX design, code efficiency, and the successful implementation of core features under high-pressure conditions.
- Collaborated in a team environment to architect a scalable frontend/backend solution, demonstrating strong problem-solving and full-stack development skills.

EDUCATION

- **Bachelors of Software Engineering** (Aug 2023 - Expected 2027)
  FAST NUCES Lahore, PK

- **FSC Pre Engineering** (June 2019 - July 2021)
  PIASS College Kasur, PK